Lab 1 – Tensile Test

Final Report

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Objective

This experiment measures the tensile behavior of aluminum and brass using a bench-top tensile test machine to determine their material properties, including Young's modulus, yield strength, ultimate tensile strength, ductility, and toughness. For each sample, stress/strain data is collected as a load is applied until fracture.

Method

Calibration

A calibration of the extensometer was performed by recording the voltage values as a function of the change in displacement of the blades. An inch micrometer was used to accurately displace the blades in increments and the voltage was recorded. By creating a linear fit of this data, we were able to form a relation between voltage and distance in inches. We did not observe any apparent signs of hysteresis in our calibration. The equation for the line from week 1 and week 2, respectively, becomes the following:

$$x = 0.0303V + 0.0172$$

$$x = 0.0731V + 0.0525$$

where x has units in inches and V has units of Volts. The rest of the instruments take measurements in meters, so a conversion is used in the MATLAB script.

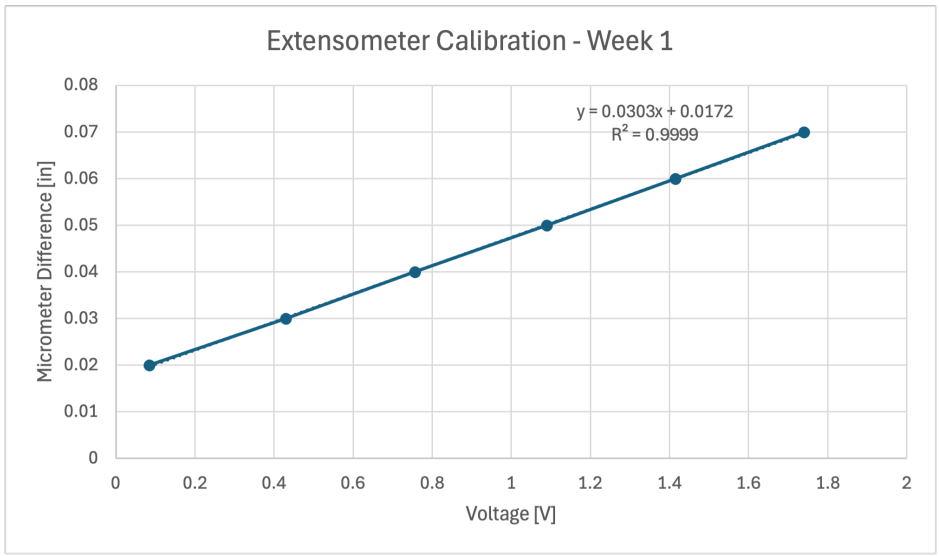


Figure 1: Extensometer Calibration for Week 1

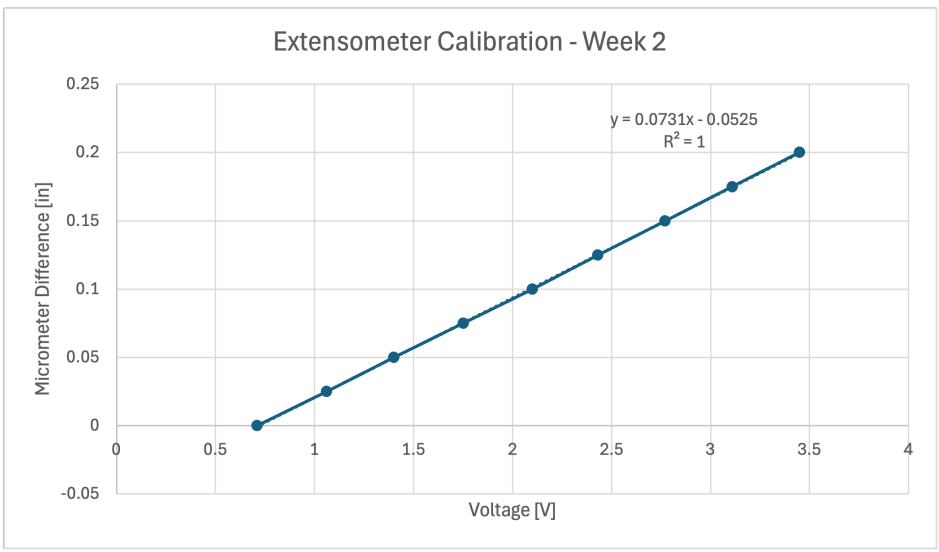


Figure 2: Extensometer Calibration for Week 2

Data

Initial Brass Sample Measurements

Sample	Length (mm)	Width (mm)	Thickness (mm)	Area (mm^2)
1	10.12	2.04	0.52	1.061
2	10.20	2.04	0.52	1.061
3	10.38	2.04	0.52	1.061
4	9.92	2.04	0.52	1.061
5	10.20	2.04	0.52	1.061

Final Brass Sample Measurements

Sample	Length (mm)	Width (mm)	Thickness (mm)	Area (mm^2)
1	12.72	1.84	0.42	0.773
2	12.50	1.82	0.41	0.746
3	12.60	1.78	0.42	0.748
4	12.30	1.82	0.42	0.764
5	12.48	1.76	0.41	0.722

Initial Aluminum Sample Measurements

Sample	Length (mm)	Width (mm)	Thickness (mm)	Area (mm^2)
1	9.80	2.04	0.52	1.061
2	9.84	2.04	0.52	1.061
3	10.20	2.04	0.52	1.061
4	9.84	2.04	0.52	1.061
5	9.91	2.04	0.53	1.061

Final Aluminum Sample Measurements

Sample	Length (mm)	Width (mm)	Thickness (mm)	Area (mm^2)
1	11.44	1.92	0.46	0.883
2	11.36	1.92	0.45	0.864
3	11.74	1.92	0.46	0.883
4	11.48	1.90	0.46	0.874
5	11.52	1.89	0.45	0.851

Initial Week 2 Sample Measurements

Sample	Length (mm)	Width (mm)	Thickness (mm)	Area (mm^2)
Unannealed Steel	10.50	2.03	0.50	1.015
Annealed Steel	9.89	2.05	0.50	1.025
Brass (Bent)	9.20	2.04	0.50	1.020

Final Week 2 Sample Measurements

Sample	Length (mm)	Width (mm)	Thickness (mm)	Area (mm^2)
Unannealed Steel	10.75	2.00	0.49	0.980
Annealed Steel	13.71	1.22	0.40	0.488
Brass (Bent)	1.020	1.77	0.40	0.708

Stress–Strain Properties for Brass Samples

Sample	Young's Modulus (GPa)	0.2% Yield Stress (MPa)	UTS (MPa)	Toughness (MJ/m^3)
1	102.02 ± 1.18	372 ± 8.32	419.94 ± 9.39	36.86 ± 2.17
2	104.10 ± 1.47	375 ± 8.38	424.08 ± 9.48	36.54 ± 1.93
3	97.92 ± 1.43	368 ± 8.23	419.80 ± 9.38	38.77 ± 1.81
4	102.94 ± 1.15	370 ± 8.27	421.25 ± 9.41	35.90 ± 2.04
5	111.63 ± 1.69	382 ± 8.54	430.42 ± 9.62	32.95 ± 1.94
Average	103.72	373	423.10	36.20

Stress–Strain Properties for Aluminum Samples

Sample	Young's Modulus (GPa)	0.2% Yield Stress (MPa)	UTS (MPa)	Toughness (MJ/m^3)
1	85.10 ± 1.54	315 ± 7.05	436.50 ± 9.75	17.40 ± 1.48
2	105.56 ± 1.01	314 ± 7.02	426.84 ± 9.54	15.14 ± 1.34
3	109.43 ± 1.00	338 ± 7.56	443.19 ± 9.90	16.93 ± 1.36
4	106.52 ± 1.13	318 ± 7.11	430.36 ± 9.62	18.75 ± 1.46
5	126.02 ± 1.63	345 ± 7.71	421.18 ± 9.41	14.72 ± 1.39
Average	106.53	326	431.61	16.59

Week 2 Samples Stress–Strain Data

Sample	Young's Modulus (GPa)	0.2% Yield Stress (MPa)	UTS (MPa)	Toughness (MJ/m^3)
Unannealed Steel	213.46 ± 0.23	884 ± 19.89	931.9 ± 20.97	25.26 ± 0.60
Annealed Steel	125.08 ± 1.52	287 ± 6.47	365.0 ± 8.22	102.47 ± 2.86
Brass (Bent)	35.78 ± 0.20	326 ± 7.35	451.9 ± 10.17	93.98 ± 2.55

Ductility Results for Brass

Sample	Ductility by Elongation at Fracture (%)	Ductility by Reduction in Area (%)
1	25.69 ± 0.05	27.15 ± 0.01
2	22.55 ± 0.05	29.66 ± 0.01
3	21.39 ± 0.05	29.52 ± 0.01
4	23.99 ± 0.05	27.94 ± 0.01
5	22.35 ± 0.05	31.98 ± 0.01
Average	23.19	29.25

Ductility Results for Aluminum

Sample	Ductility by Elongation at Fracture (%)	Ductility by Reduction in Area (%)
1	16.73 ± 0.05	16.74 ± 0.01
2	15.45 ± 0.05	18.55 ± 0.01
3	15.10 ± 0.05	16.74 ± 0.01
4	16.67 ± 0.05	17.61 ± 0.01
5	16.25 ± 0.05	19.82 ± 0.01
Average	16.04	17.89

Ductility Results for Week 2 Samples

Sample	Ductility by Elongation at Fracture (%)	Ductility by Reduction in Area (%)
Unannealed Steel	2.38 ± 0.05	3.45 ± 0.01
Annealed Steel	38.62 ± 0.05	52.39 ± 0.01
Brass (Bent)	27.72 ± 0.05	30.59 ± 0.01

Analysis

The values for Young's Modulus, UTS, and Toughness were all calculated using the MATLAB script in the Appendix, while the 0.2% Yield Stress was determined graphically. The uncertainty values were also calculated using MATLAB, as referenced in the Appendix.

Calculations

Calculations and Uncertainty Propagation

Ductility — Reduction of Area (RA):

$$\mathrm{RA} = \frac{\text{original cross sectional area} - \text{minimum final area}}{\text{original cross sectional area}} = \frac{A_0 - A_{\min}}{A_0} = 1 - \frac{w_f t_f}{w_0 t_0}.$$

Uncertainty in Reduction of Area:

$$\Delta(\mathrm{RA}) = \sqrt{\left(\frac{\partial \mathrm{RA}}{\partial t_f}\Delta t\right)^2 + \left(\frac{\partial \mathrm{RA}}{\partial w_f}\Delta w\right)^2 + \left(\frac{\partial \mathrm{RA}}{\partial t_0}\Delta t\right)^2 + \left(\frac{\partial \mathrm{RA}}{\partial w_0}\Delta w\right)^2},$$

where

$$\frac{\partial \mathrm{RA}}{\partial t_f} = -\frac{w_f}{w_0 t_0}, \quad \frac{\partial \mathrm{RA}}{\partial w_f} = -\frac{t_f}{w_0 t_0}, \quad \frac{\partial \mathrm{RA}}{\partial t_0} = \frac{w_f t_f}{w_0 t_0^2}, \quad \frac{\partial \mathrm{RA}}{\partial w_0} = \frac{w_f t_f}{w_0^2 t_0}.$$

Ductility — Elongation at Fracture:

$$G = \frac{\text{Final gauge length} - \text{Original gauge length}}{\text{Original gauge length}} = \frac{L_f - L_0}{L_0} \qquad \left[\,\%\,\text{elongation} = 100 \times \frac{L_f - L_0}{L_0}\,\right]$$

Uncertainty in Elongation:

$$\Delta G = 100\,\frac{\Delta L}{L_0},$$

where ΔL is the caliper uncertainty on L_f.

Toughness (area under stress–strain curve performed via MATLAB):

$$T = \int \sigma\, d\varepsilon \approx \sigma \times \varepsilon.$$

Uncertainty in Toughness (approximation):

$$\Delta T = \sqrt{\left(\sigma_f\,\tfrac{\Delta L}{L_0}\right)^2 + \left(\varepsilon_f\,\tfrac{\sigma_f}{m_f}\Delta m_f\right)^2 + \left(\varepsilon_f\,\tfrac{m_f}{wt}\Delta V\right)^2 + \left(\varepsilon_f\,\tfrac{\sigma_f}{w}\Delta w\right)^2 + \left(\varepsilon_f\,\tfrac{\sigma_f}{t}\Delta t\right)^2}.$$

Young's Modulus:

$$\varepsilon_i = \frac{\Delta L_i}{L_0}, \qquad \sigma_i = \frac{F_i}{wt}.$$

Restricting to the linear region:

$$\sigma \approx m\,\varepsilon + b, \quad \text{so} \quad E = m.$$

Linear Fit Uncertainty:

$$\Delta m = s\sqrt{\frac{n}{n\sum_i x_i^2 - \left(\sum_i x_i\right)^2}} \qquad \Delta b = s\sqrt{\frac{\sum_i x_i^2}{n\sum_i x_i^2 - \left(\sum_i x_i\right)^2}},$$

where

$$s = \sqrt{\frac{\sum_i y_i^2 - \frac{1}{n}\left(\sum_i y_i\right)^2 - m\left(\sum_i x_i y_i - \frac{1}{n}\sum_i x_i \sum_i y_i\right)}{n-2}}.$$

Ultimate Tensile Stress (UTS):

$$\sigma_{\text{UTS}} = \frac{m_f\, V_{\max}}{w\, t}.$$

with

$$m_f = 444.82 \text{ N/V}, \qquad \Delta m_f = 0.02\, m_f, \tag{1}$$

$$V_{\max} = \max(V) \text{ [V]}, \qquad \Delta V = 0.001 \text{ V}, \tag{2}$$

$$w = 2.04 \times 10^{-3} \text{ m}, \qquad \Delta w = 0.005 \times 10^{-3} \text{ m}, \tag{3}$$

$$t = 0.52 \times 10^{-3} \text{ m}, \qquad \Delta t = 0.005 \times 10^{-3} \text{ m}. \tag{4}$$

The propagated relative uncertainty is

$$\frac{\Delta\sigma}{\sigma} = \sqrt{r_{m_f}^2 + r_V^2 + r_w^2 + r_t^2},$$

$$\Delta\sigma_{\text{UTS}} = \sigma_{\text{UTS}}\sqrt{r_{m_f}^2 + r_V^2 + r_w^2 + r_t^2}.$$

0.2% Offset Yield Stress:

$$\sigma_{0.2} = \frac{m_f V_{0.2}}{wt}.$$

Uncertainty in 0.2% Offset Yield Stress:

$$\Delta\sigma_{0.2} = \sqrt{\left(\frac{V_{0.2}}{wt}\Delta m_f\right)^2 + \left(\frac{m_f}{wt}\Delta V\right)^2 + \left(\frac{m_f V_{0.2}}{w^2 t}\Delta w\right)^2 + \left(\frac{m_f V_{0.2}}{wt^2}\Delta t\right)^2}.$$

Results

Representative Stress–Strain Graphs

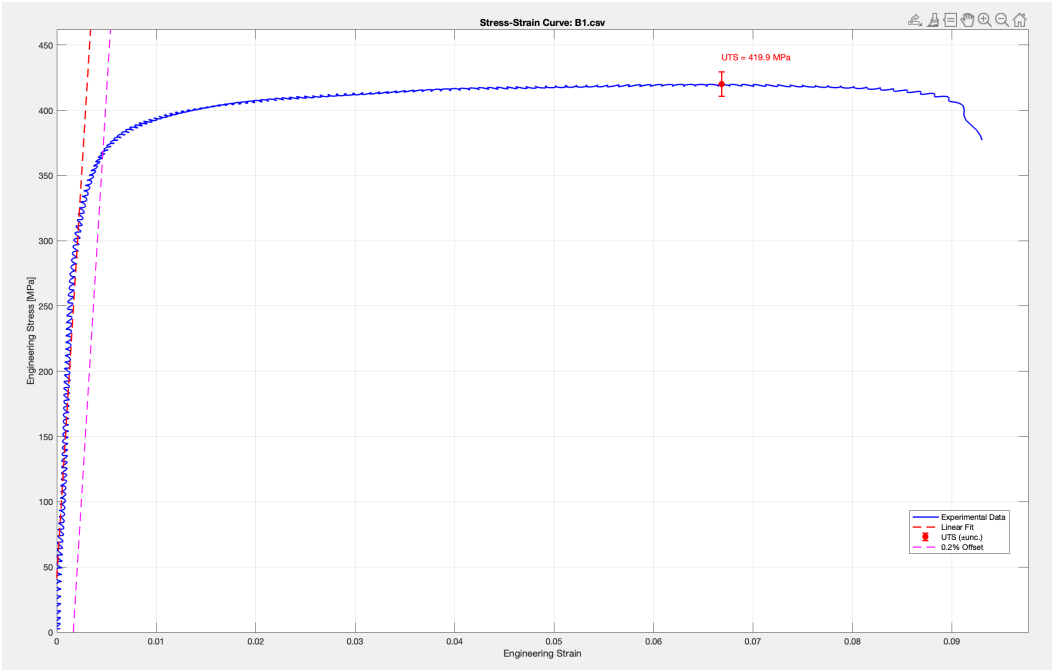


Figure 3: Brass Sample 1 Stress-Strain Curve

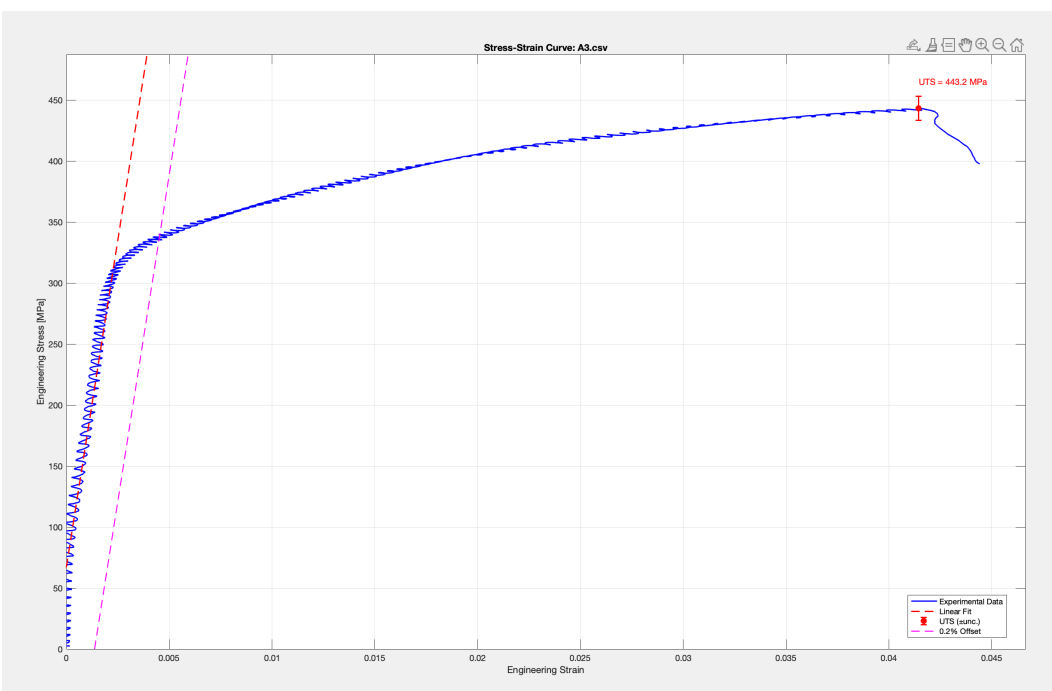


Figure 4: Aluminum Sample 3 Stress-Strain Curve

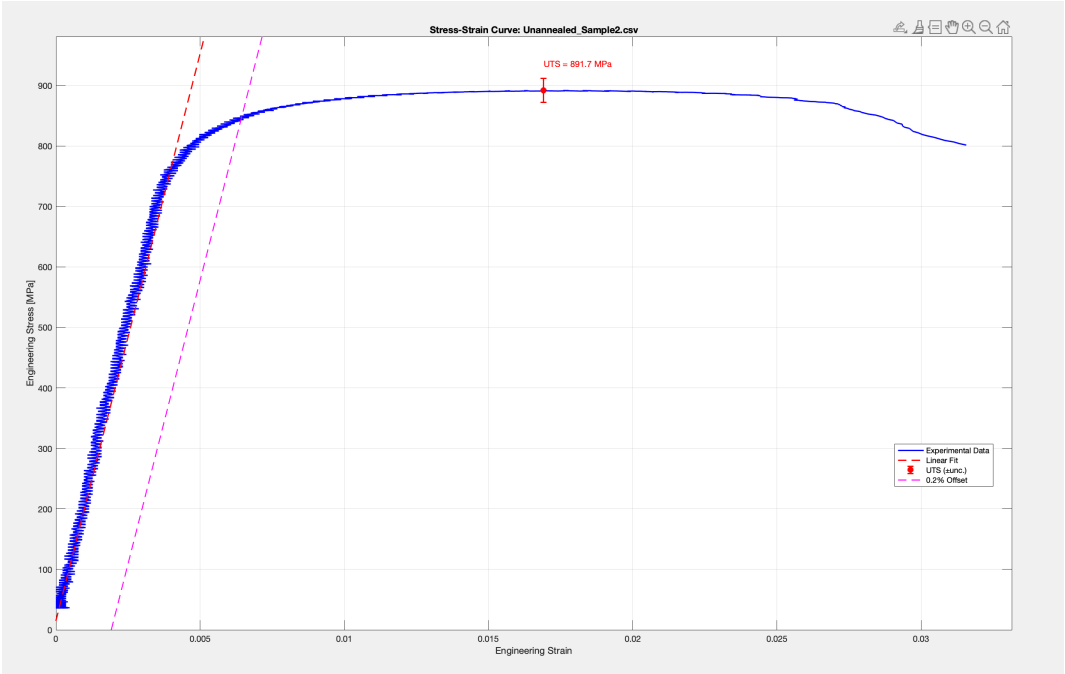


Figure 5: Unannealed Steel Sample Stress-Strain Curve

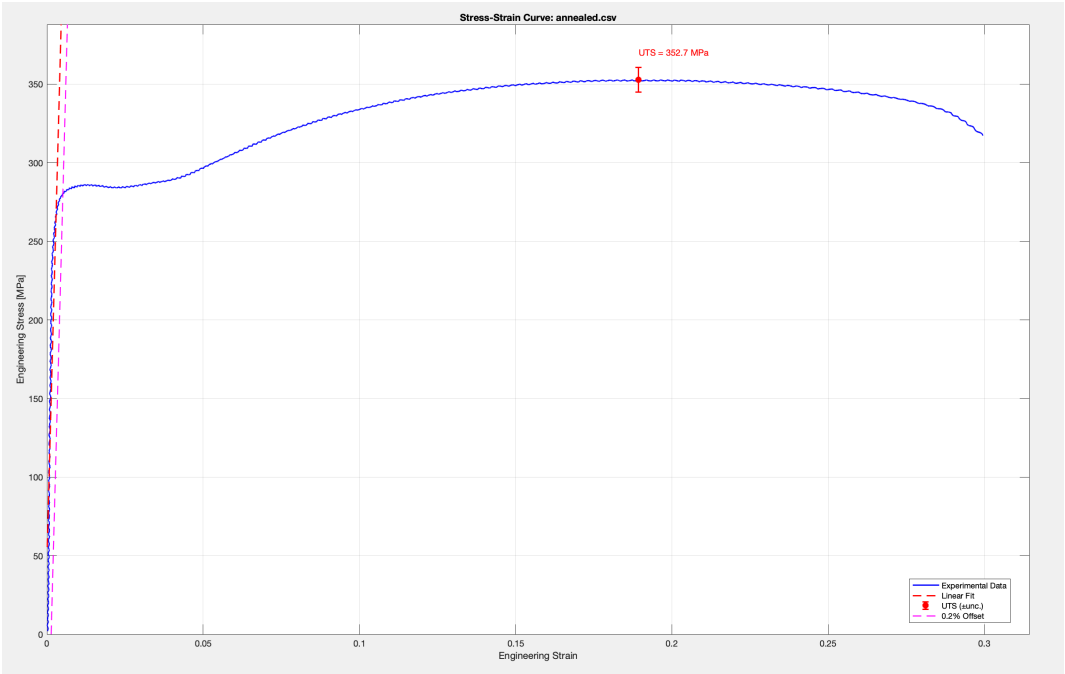


Figure 6: Annealed Steel Sample Stress-Strain Curve

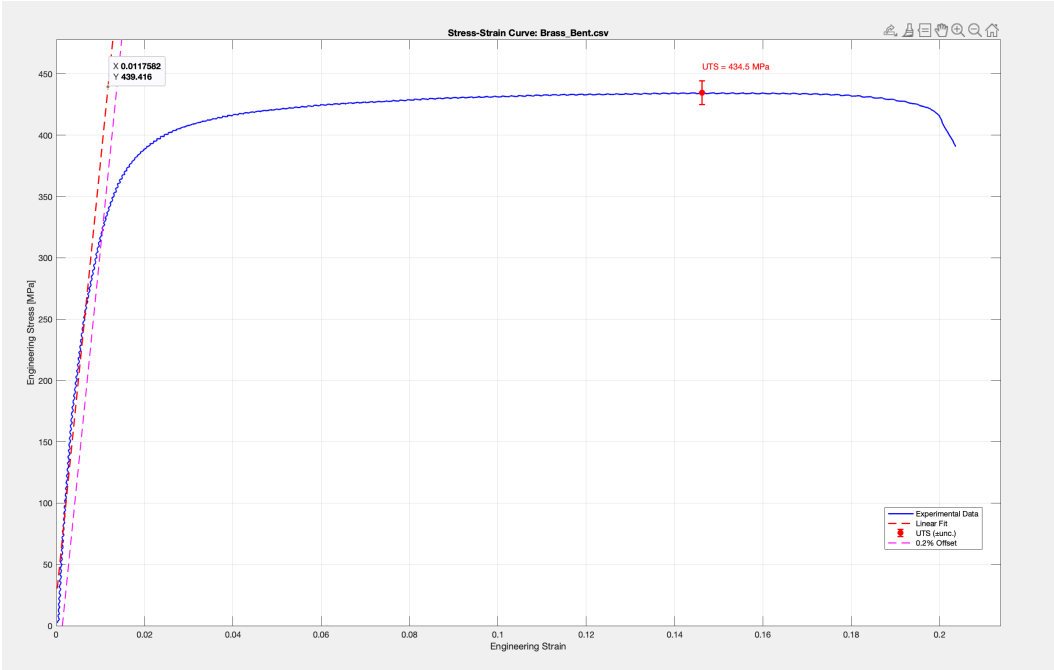
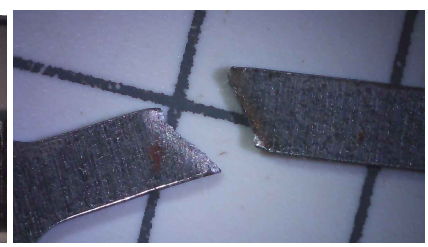


Figure 7: Brass (Bent) Sample Stress-Strain Curve

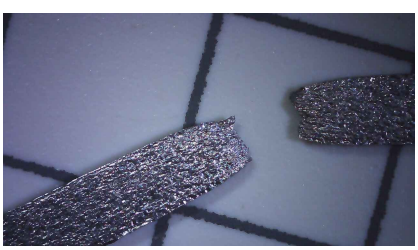
Fractured Sample Images

Fractured Brass Sample

Fractured Aluminum Sample



Fractured Unannealed Steel Sample



Fractured Annealed Steel Sample

Pre-Fracture Bent Brass Sample

Fractured Bent Brass Sample

Discussion

By examining the results of this experiment, it is apparent that the material properties of different materials greatly effect how they perform under stress. Qualitatively, the cross-sectional area of the samples was reduced more in the samples with higher ductility, increasing even more when bent to improve the reaction to stretching. This makes sense, as higher ductility meant that it could stretch more before being torn apart. The break in the annealed sample as well as in both brass samples showed more stretching and necking, while the aluminum and unannealed steel had a very sharp edge with minimal necking.

The values attained by both sets of samples start to get a little confusing, as brass should have a higher Young's Modulus, Ultimate Tensile Strength, Toughness, and 0.2% Yield Strength. However, when comparing the average results, the average ultimate tensile strength of the aluminum samples (431.61 MPa) was higher than that of brass (423.1 MPa). When comparing this to the values of unannealed steel (931.9 MPa), annealed steel (365 MPa), and the bent brass (451.9 MPa) it is clear that the softer metals have a lower 0.2% yield stress. Aluminum had an average Young's Modulus of 106.53 GPa, while the unbent brass samples were found to have 103.72 GPa. The unannealed steel had a value of 213.46 GPa, annealed steel had a value of 125.08 GPa, and the bent brass had a value of 35.78 GPa. These values make sense as the lower values demonstrated much higher elongation at failure than those with higher values which had much more clean breaks.

When observing the values of the 0.2% Yield Strength, aluminum had an average value of 326 MPa while brass had an average of 373 MPa. The week two samples had the following values: unannealed steel, 884 MPa; annealed steel, 287; and bent brass, 326 MPa. Lastly is the comparison of toughness. Aluminum had an average value of 16.59 MJ/m^3 and unbent brass had a toughness of 36.20 MJ/m^3. Unannealed steel had a toughness of 25.26 MJ/m^3, annealed steel a value of 102.47 MJ/m^3, and bent brass with a value of 93.98 MJ/m^3. These values make sense as the area under the stress strain curve increased a lot more when there was more necking before failure allowing for larger values.

wNow, when comparing ductility values, brass had average values of 23.19% ductility by elongation at failure and 29.25% by reduction in area.. The aluminum samples had average values of 16.04% ductility by elongation at failure and 17.89% by reduction in area. The ductility of unannealed steel was 2.38% by elongation and 3.45% by reduction in area. Annealed steel sported a value of 38.62% by elongation and 52.39 by reduction in area. Bent brass, similar to annealed steel, had values of 37.72% and 30.59% respectively. These all seem reasonable when compared to one another. They are unrelated to the calibration performed on the extensometer, which is where we believe there to be systematic error that caused the inconsistencies in our data to be found.

The annealing process typically lowers $\sigma_{0.2}$ and the UTS while increasing the materials elongation which matches our findings where unannealed steel had a UTS of 287 MPa while unannealed had a UTS of 931.9 MPa. In this lab we also tested cold working and how it affects brass samples.

Cold working or pre-bending is typically used to increase the UTS which is what we found in our results. Cold working the brass samples increased the UTS from 423 MPa to 451 MPa.

Visual inspection of the broken samples shows that the annealed steel sample is the only sample to experience a ductile fracture. The sample experienced lots of necking and the final cross sectional area of the sample is small when compared to the other samples. This can be seen in the stress-strain plot where there is a region of necking and the stress proceeds to increase further. In all other samples, the curve appears to continue in a steady curve while the annealed sample is clearly different. The aluminum curve continued to increase to a point of failure while the remaining three samples (both brasses and unannealed steel) seem to plateau at a certain stress, then drop suddenly. All other samples showed signs of brittle fractures where they did not experience much necking and the final cross sectional ares that we measured support this. We do expect a large error in the final cross sectional areas due to how hard it was to measure them with calipers. Our results are a rough estimation.

In the first half of the lab, the extensometer was in fine working condition, and the data collected (for the brass samples) was found to be consistent with expected results. It was only in the second half that the data began to provide unexpected results. We believe this was caused by the extensometer becoming damaged in some way. The teeth were able to rotate much more freely than before, causing much more "play" in the blades. This made mounting the extensometer difficult and subject to error. This could have rendered our original calibration useless, giving us inaccurate data for the aluminum samples.

Additionally, inconsistencies in the data are due in part because of our lab situation. Our team had a non-functioning extensometer, so we were split into three different groups. We joined each group late into the calibration process and data collection, and some information seemed to be missing from all three of our groups, but for various reasons. We used the most complete data we could find, but overall, it may not have been the "best." The disorganized nature of having three different lab experiences made the analysis very difficult.

Appendix — MATLAB

Tensile Test Analysis Script

```
%% Tensile Test Analysis Script with All Uncertainties
% Computes stress-strain behavior, modulus, UTS, ductility, toughness,
% and uncertainty propagation for all key parameters.

clear; clc; close all;

%% USER INPUTS
filename = 'A5.csv';

% Initial dimensions
width0 = 2.04e-3;
thickness0 = 0.52e-3;
gauge_length0 = 9.91e-3;

% Post-test dimensions
L_final = 11.52e-3;
width_final = 1.89e-3;
thickness_final = 0.45e-3;

% Linear region for modulus
linear_strain_max = 0.003;

% Floor trimming threshold
floor_fraction = 0.005;

% 0.2% offset
offset = 0.002;

% Load cell & DAQ
m_f = 444.82;
rel_unc_loadcell = 0.02;
Delta_mf = rel_unc_loadcell * m_f;
DeltaV = 0.001;

% Caliper uncertainties
Delta_w = 0.005e-3;
Delta_t = 0.005e-3;

%% READ DATA
data = readmatrix(filename);
time = data(:,1);
```

```
displacementVolt = data(:,2);
displacement = ((displacementVolt * 0.0303) - 0.0172) / 39.37; % meters
displacement = displacement - mean(displacement(1:10));
load_V = data(:,3);
load_N = m_f * load_V;

%% ENGINEERING STRESS & STRAIN
A0 = width0 * thickness0;
strain = displacement / gauge_length0;
stress = load_N / A0;

%% TRIM FLOOR REGION
floor_threshold = floor_fraction * max(stress);
valid_idx = find(stress > floor_threshold, 1, 'first');
strain_trimmed = strain(valid_idx:end);
stress_trimmed = stress(valid_idx:end);

%% ULTIMATE TENSILE STRESS
[UTS, idx_uts] = max(stress_trimmed);
strain_uts = strain_trimmed(idx_uts);

%% TRIM POST-FRACTURE DATA
drop_threshold = 0.9;
idx_drop = find(stress_trimmed(idx_uts:end) < drop_threshold*UTS, 1, 'first');
if ~isempty(idx_drop)
    fracture_idx = idx_uts + idx_drop - 1;
    strain_trimmed = strain_trimmed(1:fracture_idx);
    stress_trimmed = stress_trimmed(1:fracture_idx);
end

%% YOUNG'S MODULUS
linear_region = strain_trimmed <= linear_strain_max;
x = strain_trimmed(linear_region);
y = stress_trimmed(linear_region);

p = polyfit(x, y, 1);
E = p(1);

% Linear fit uncertainty
n = length(x);
Sx = sum(x);
Sy = sum(y);
Sxx = sum(x.^2);
Sxy = sum(x .* y);
```

```

m = (n*Sxy - Sx*Sy)/(n*Sxx - Sx^2);
b = (Sy - m*Sx)/n;

y_fit = m*x + b;
residuals = y - y_fit;
s = sqrt(sum(residuals.^2)/(n-2));
dm = s * sqrt(n/(n*Sxx - Sx^2));

%% DUCTILITY
ductility_elongation = ((L_final - gauge_length0) / gauge_length0) * 100;
delta_L = 0.005e-3;
delta_elongation = delta_L / gauge_length0 * 100;

A_final = width_final * thickness_final;
ductility_RA = ((A0 - A_final)/A0)*100;

dRAdt  = (-width_final) / (thickness0 * width0);
dRAdw  = (-thickness_final) / (thickness0 * width0);
dRAdt0 = (thickness_final*width_final) / (width0 * thickness0^2);
dRAdw0 = (thickness_final*width_final) / (width0^2 * thickness0);
delta_RA = sqrt((dRAdt * Delta_t)^2 + (dRAdw * Delta_w)^2 + ...
                (dRAdt0 * Delta_t)^2 + (dRAdw0 * Delta_w)^2);

%% TOUGHNESS
Toughness = trapz(strain_trimmed, stress_trimmed);
stress_final = stress_trimmed(end);
strain_final = (L_final - gauge_length0)/gauge_length0;

% Toughness uncertainty (approx.)
dT_dstrain = stress_final;
dT_dmf     = strain_final * (stress_final / m_f);
dT_dV      = strain_final * (m_f / (width0*thickness0));
dT_dw      = strain_final * (-stress_final / width0);
dT_dt      = strain_final * (-stress_final / thickness0);

delta_T = sqrt( ...
    (dT_dstrain * delta_L / gauge_length0)^2 + ...
    (dT_dmf * Delta_mf)^2 + ...
    (dT_dV * DeltaV)^2 + ...
    (dT_dw * Delta_w)^2 + ...
    (dT_dt * Delta_t)^2 );

%% UTS UNCERTAINTY
```

```
Vmax = max(load_V);
sigma_uts = (m_f * Vmax) / (width0 * thickness0);

rel_mf = Delta_mf / m_f;
rel_V  = DeltaV / Vmax;
rel_w  = Delta_w / width0;
rel_t  = Delta_t / thickness0;
rel_sigma = sqrt(rel_mf^2 + rel_V^2 + rel_w^2 + rel_t^2);
Delta_sigma_UTS = sigma_uts * rel_sigma;

%% PLOT FIRST
figure;
plot(strain_trimmed, stress_trimmed/1e6, 'b', 'LineWidth', 1.5); hold on;

strain_fit_plot = linspace(min(strain_trimmed), 0.8*max(strain_trimmed), 100);
stress_fit_plot = polyval(p, strain_fit_plot);
plot(strain_fit_plot, stress_fit_plot/1e6, 'r--', 'LineWidth', 1.5);

plot(strain_uts, UTS/1e6, 'ro', 'MarkerSize', 8, 'MarkerFaceColor','r');
text(strain_uts, UTS/1e6*1.05, sprintf('UTS = %.1f MPa', UTS/1e6), 'Color','r');

stress_offset_plot = polyval(p, strain_fit_plot - offset);
plot(strain_fit_plot, stress_offset_plot/1e6, 'm--', 'LineWidth', 1.2);
text(strain_fit_plot(end)*0.7, stress_offset_plot(end)/1e6, '0.2% Offset', 'Color','m');

xlabel('Engineering Strain');
ylabel('Engineering Stress [MPa]');
title(['Stress-Strain Curve: ' filename], 'Interpreter','none');
grid on;
xlim([0 max(strain_trimmed)*1.05]);
ylim([0 max(stress_trimmed/1e6)*1.1]);
legend('Experimental Data', 'Linear Fit', 'UTS', '0.2% Offset', 'Location','best');

%% AFTER PLOT: USER INPUT for 0.2% Yield
stress_02_MPa = input('\nEnter the 0.2%% offset yield stress [MPa]: ');
stress_02 = stress_02_MPa * 1e6;  % convert to Pa

% Backsolve voltage
V_02 = stress_02 * width0 * thickness0 / m_f;

% Partial derivatives for propagation
dsigma_dmf = V_02 / (width0 * thickness0);
dsigma_dV  = m_f / (width0 * thickness0);
dsigma_dw  = (-m_f * V_02) / (width0^2 * thickness0);
```

```
dsigma_dt  = (-m_f * V_02) / (width0 * thickness0^2);

delta_sigma02 = sqrt((dsigma_dmf * Delta_mf)^2 + ...
                     (dsigma_dV * DeltaV)^2 + ...
                     (dsigma_dw * Delta_w)^2 + ...
                     (dsigma_dt * Delta_t)^2);

%% PRINT RESULTS
fprintf('\n=== RESULTS ===\n');
fprintf('UTS: %.2f  %.2f MPa\n', UTS/1e6, Delta_sigma_UTS/1e6);
fprintf('Young''s Modulus: %.2f  %.2f GPa\n', E/1e9, dm/1e9);
fprintf('Elongation at Fracture: %.2f  %.2f %%\n', ductility_elongation, delta_elongation
    );
fprintf('Reduction of Area: %.2f  %.2f %%\n', ductility_RA, delta_RA);
fprintf('Toughness: %.2f  %.2f MJ/m^3\n', Toughness/1e6, delta_T/1e6);
fprintf('0.2%% Offset Yield Stress: %.2f  %.2f MPa\n', stress_02/1e6, delta_sigma02/1e6);
```